



BRIEFING FOR NEW BUILD DEVELOPERS

Meeting the Future Homes Standard without an outdoor unit on every plot

How the Hestora thermal energy store gives housebuilders a compliant, buyer-friendly heating option that pairs with mandated rooftop solar and adds nothing to the evening peak.

Hestora Limited · August 2026 · www.hestora.net

1. The Future Homes Standard changes your heating decision

The Future Homes Standard (Part L 2026) ends fossil-fuel heating in new homes in England. New residential submissions must comply from **24 March 2027**, with unstarted plots caught from March 2028. Compliance is assessed under SAP 10.3 (with the Home Energy Model to follow), targeting a ~75–80% reduction in carbon emissions against the 2013 baseline. Gas and oil systems are now the least viable compliance route, and rooftop solar equivalent to roughly 40% of ground-floor area is expected on most new homes.

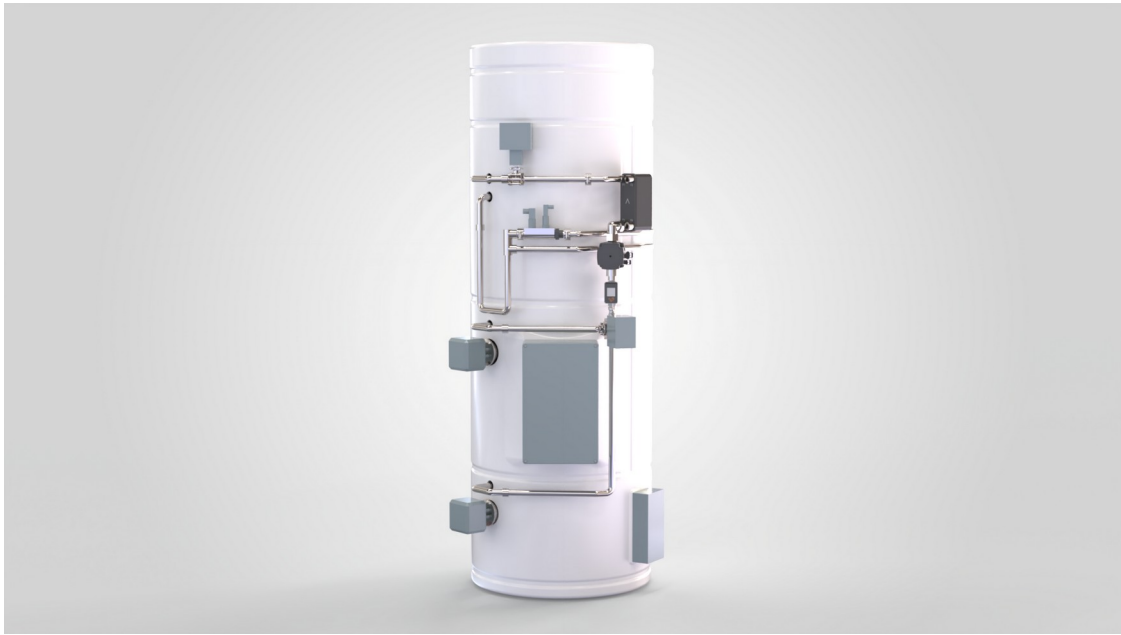
Date	What happens
24 March 2027	New residential dwelling submissions in England must comply with the FHS
24 September 2027	Higher-risk (tall residential) buildings must adopt the updated standards
24 March 2028	Transitional cut-off — unstarted plots must comply

For most developers the default answer is an air-source heat pump on every plot. That works — but it brings an outdoor unit into every garden or balcony design, noise and separation constraints into planning, emitter and cylinder specification into every house type, a stretched installer skills base into your programme, and a new electrical evening peak into your grid connection.

2. A second compliant route: store the energy instead

Hestora is a 40 kWh pressurised, water-based thermal energy store that installs inside the home, occupying the space of a hot-water cylinder: 650 mm × 650 mm and 1,800 mm tall. Because it arrives light and dry, it can be sited anywhere in the dwelling — plant cupboard, utility, garage or loft. It charges from cheap off-peak electricity overnight — and from the home's own mandated solar array during the day — then delivers a full day of space heating and mains-pressure hot water through a conventional wet system. Zero

emissions at the plot, no refrigerants, no outdoor unit, and a >30-year design life built on water, steel and insulation.



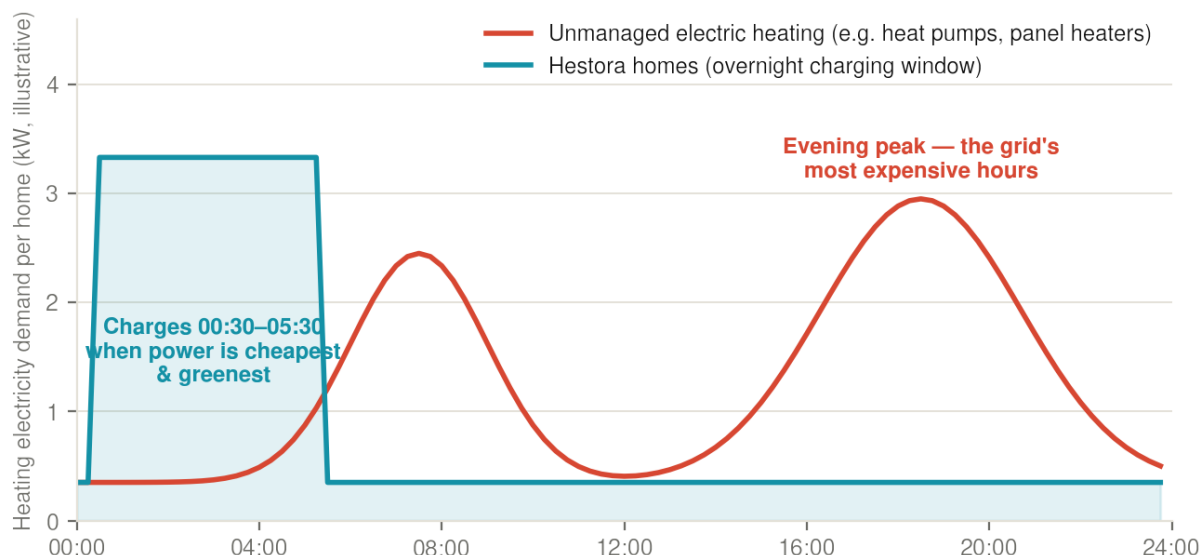
The Hestora MKII. Everything lives inside the envelope: pressure vessel, insulation, heat exchangers and smart controls — nothing on the wall outside.

- **Design freedom** — no outdoor unit means no acoustic separation distances, no visual screening, no lost garden or balcony space, and cleaner elevations on dense sites and urban infill.
- **Simpler specification** — standard wet central heating at boiler-like flow temperatures: conventional radiators sized normally, or underfloor heating at low temperatures — either works, with no oversizing exercise per house type.
- **Programme-friendly** — the unit arrives light and dry, moves through a finished house easily, is filled on site and commissioned in around a day, by wet-trade skills your supply chain already has.
- **Pairs with mandated solar** — the FHS puts PV on the roof; Hestora gives that generation somewhere to go. A built-in diverter charges the store from surplus solar first, and depending on the size of the array this can mean virtually no heating and hot-water costs for the occupant from April to October — a running-cost story no gas-era home could tell.

3. The grid connection argument

Every all-electric development lives or dies by its network connection. Unmanaged electric heating — heat pumps included — draws hardest on winter evenings, exactly when the local network is most stressed; at scale that can mean reinforcement costs or even a new substation on the development. Hestora consumes nothing at the peak: it charges in the small hours and coasts through the day on stored heat.

When homes draw power — unmanaged electric heating vs Hestora



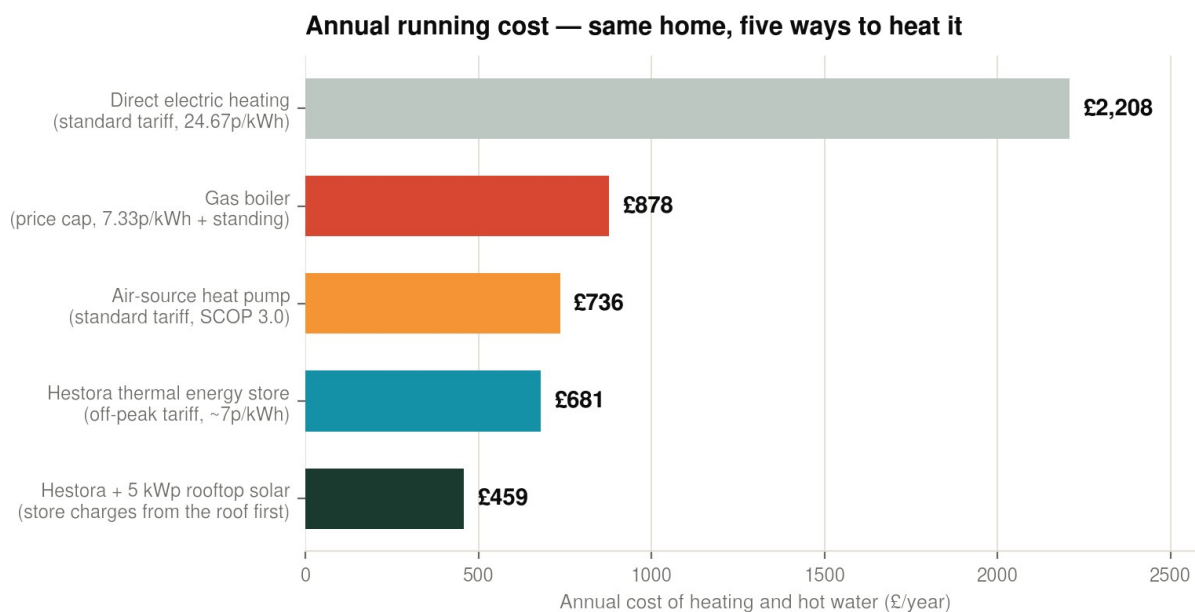
Illustrative winter-day heating electricity demand per home. Unmanaged electric heating coincides with the network's morning and evening peaks; Hestora concentrates its draw in the 00:30–05:30 window and is near-silent thereafter. Diversified across a development, that difference materially changes the connection-capacity conversation with the DNO.

Per-plot demand profile

Each unit absorbs up to 40 kWh overnight. Phase 1 charging uses the off-peak window; from mid-2027 units will additionally integrate with NESO's Demand Flexibility Service 'turn-up' mechanism, which pays aggregated controllable demand to absorb surplus renewables. A fleet of Hestora homes is a grid asset, not a grid problem.

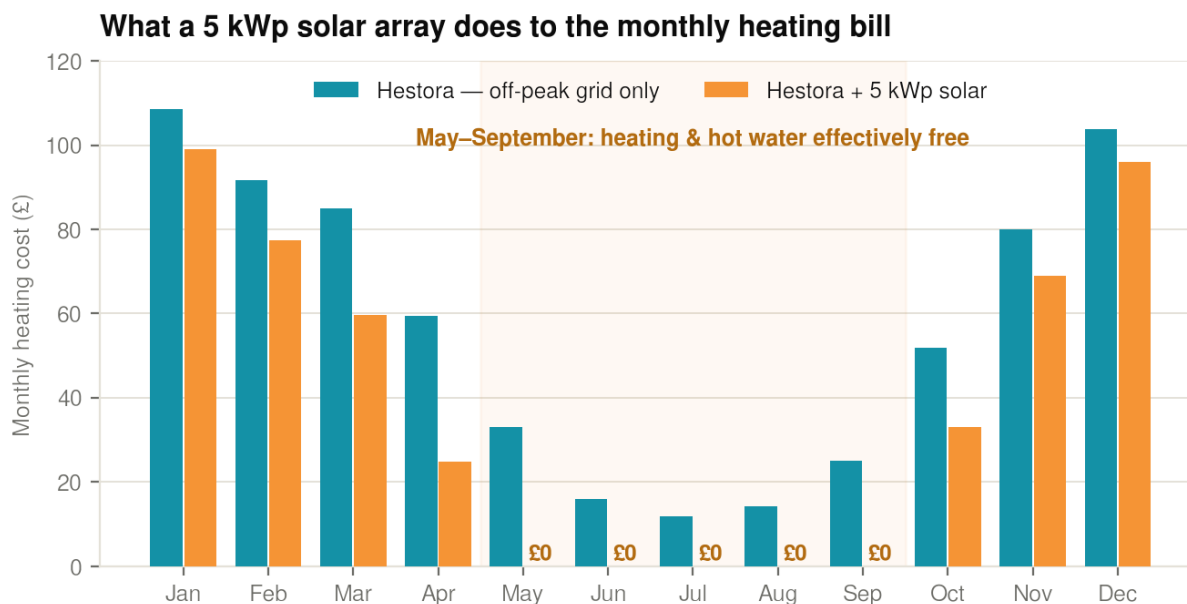
4. The buyer's story

FHS homes will be sold partly on running costs. Our modelling — built on a living-lab dataset of ~6.4 million half-hourly smart-meter readings from 200 British homes — shows what the heating bill looks like in a typical home.



Modelled annual cost of meeting a typical existing home's 8,950 kWh heat and hot-water demand, mid-2026 prices. Assumptions: direct electric at the standard single rate (24.67p/kWh); gas at the current cap (7.33p/kWh plus standing charge, 85% boiler efficiency); air-source heat pump at SCOP 3.0, spanning £631–£788 across the plausible 3.5–2.8 range; Hestora at ~7p/kWh off-peak, and with a 5 kWp array feeding the store first. An FHS new build's demand may be half this, shrinking every bar — but the ranking holds, and the all-electric home sheds the gas standing charge.

The FHS already puts the array on the roof — the chart below shows what that mandated solar is worth when the store charges from it first. On a typical demand profile, a 5 kWp array makes the home's heating and hot water effectively free from May to September:



Monthly heating and hot-water cost with Hestora charged from off-peak grid electricity alone (blue) and with a 5 kWp south-facing array feeding the store ahead of the grid (amber). May–September falls to zero and the annual cost drops from ~£681 to ~£459. Modelled against existing-stock demand: an FHS new build's lower demand means a given array covers proportionally more, and specifying 6–7 kWp takes April and October to zero as well.

- **A warm-home experience buyers recognise** — mains-pressure hot water and radiator heat indistinguishable from a good boiler, with nothing to learn and nothing outside to look at or listen to.

- **Future-proof by design** — as the grid decarbonises, the carbon intensity of stored heat falls automatically — the home's emissions story improves every year without further work.

5. Status and how to engage

Hestora is a certified full-size prototype: the pressure vessel is independently certified to PD 5500:2024+A2 (Category 2) and tested to 17.86 bar — more than double its 8 bar operating pressure — by British Engineering Services. A monitored pilot begins in November 2026; MCS certification — which depends on the BSI standard for thermal energy storage being defined and testing by BRE, awaited by our whole category — follows; commercial sales open in 2027, aligned, usefully, with the FHS compliance dates above.

We are looking for developer partners for early plots and show homes: sites where an indoor, solar-coupled, peak-free heating system solves a specific design, planning or connection problem.

Next steps

Visit www.hestora.net to see the product in action, or contact jim@hestora.net to discuss your pipeline. Hestora Limited is a British engineering company based in the Midlands, supported by energy and enterprise programmes at Imperial College London and the University of Birmingham.